

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P2-Model)** **Date: 10-05-2026**

Time: 02.30Pm to 05.30Pm **GTA-31** **Max. Marks: 180**

10-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2023-P2)_GTA-31_Syllabus

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2023-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N :5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N :8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N :14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N :18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N :25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N :31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N :35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N :42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N :48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. $\vec{a},\vec{b},\vec{c}$ are three non-zero coplanar vectors. If $\vec{a}$ is not parallel to $\vec{b}$, then $\vec{c}=$

A) $\dfrac{\begin{vmatrix}\vec{a}.\vec{c} & \vec{a}.\vec{b}\\ \vec{c}.\vec{b} & \vec{b}.\vec{b}\end{vmatrix}\vec{a}+\begin{vmatrix}\vec{a}.\vec{a} & \vec{c}.\vec{a}\\ \vec{a}.\vec{b} & \vec{c}.\vec{b}\end{vmatrix}\vec{b}}{\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b}\\ \vec{a}.\vec{b} & \vec{b}.\vec{b}\end{vmatrix}}$

B) $\begin{vmatrix}\vec{c}.\vec{a} & \vec{a}.\vec{b}\\ \vec{c}.\vec{b} & \vec{b}.\vec{b}\end{vmatrix}\vec{a}+\begin{vmatrix}\vec{a}.\vec{a} & \vec{c}.\vec{a}\\ \vec{a}.\vec{b} & \vec{c}.\vec{b}\end{vmatrix}\vec{b}$

C) $\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b}\\ \vec{a}.\vec{b} & \vec{b}.\vec{b}\end{vmatrix}\vec{a}+\begin{vmatrix}\vec{c}.\vec{a} & \vec{a}.\vec{b}\\ \vec{c}.\vec{b} & \vec{c}.\vec{b}\end{vmatrix}\vec{b}$

D) $\dfrac{\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b}\\ \vec{a}.\vec{b} & \vec{b}.\vec{b}\end{vmatrix}\vec{a}+\begin{vmatrix}\vec{c}.\vec{a} & \vec{a}.\vec{b}\\ \vec{c}.\vec{b} & \vec{c}.\vec{b}\end{vmatrix}\vec{b}}{\begin{vmatrix}\vec{a}.\vec{a} & \vec{b}.\vec{b}\\ \vec{c}.\vec{c} & \vec{b}.\vec{b}\end{vmatrix}}$

2. PQ is a chord of the parabola of $y^2=4ax$ whose focus is at S, such that perimeter of the triangle $PQS=7$. The tangents, drawn to the parabola at $P\&Q$ intersect at T. The line joining S,T and chord PQ intersects at R. Given that the slopes of PQ and ST are $\frac{1}{2}\&-\frac{1}{2}$ respectively and the length of $TR=\frac{1}{2}$. Then the relation between the lengths $SP\ \&\ SQ$ is $(SP>SQ)$.

A) $12SP-16SQ=(SP)^2-2(SQ)^2$ **B)** $12SP-14SQ=(SP)^2-2(SQ)^2$

C) $6SP-7SQ=(SP)^2-(SQ)^2$ **D)** $6SP-8SQ=(SP)^2-(SQ)^2$

3. Let $A=\left[a_{ij}\right]_{3\times3}$ be a matric where $a_{ij}\in\{0,1\}\ \forall 1\le i,j\le 3$. Let $s_i=\sum_{k=1}^{3}a_{ik}$ and $t_j=\sum_{k=1}^{3}a_{kj}$ then the probability that the product $(s_1s_2s_3t_1t_2t_3)$ is an odd integer, is

A) $\frac{1}{2}$ **B)** $\frac{1}{8}$ **C)** $\frac{1}{16}$ **D)** $\frac{1}{32}$

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4. Solve the equation $4^{x+1.5}+9^{x+0.5}=10\cdot 6^{x}$.

A) $x=\log_{1.5}\left(\frac{4}{3}\right),\log_{1.5}2$ **B)** $x=\log_{3}\left(\frac{4}{3}\right),\log_{3}2$

C) $x=\log_{0.5}\left(\frac{4}{3}\right),\log_{0.5}2$ **D)** none of these

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. For any triangle ABC, if BC = a, CA = b, AB = c and the lengths of the medians through vertices A, B, C are x, y, z respectively and $\begin{bmatrix}a^2\\b^2\\c^2\end{bmatrix}=M\begin{bmatrix}x^2\\y^2\\z^2\end{bmatrix}$ for a certain 3 x 3 matrix M.

Then

A) Sum of elements in the principal diagonal of M is $\frac{-4}{3}$

B) M is symmetric matrix

C) Determinant of M is $\frac{64}{81}$

D) Sum of all elements of M is 4

6. Let $f(x)$ be twice differentiable function such that $f''(x)>0$ in $[0, 2]$. Then

A) $f(0)+f(2)=2f(c)$, for atleast one $c, c\in(0,2)$

B) $f(0)+f(2)<2f(1)$

C) $f(0)+f(2)>2f(1)$

D) $2f(0)+f(2)>3f\left(\frac{2}{3}\right)$

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7. Let ' S_1 ' be the set of all real points lying on $\left(y^2-4x\right)\left(x^2+y^2-4x-4y\right)=0$ and let ' S_2' ' be the set of all real points lying on $y=a(x+2)$. Then which of the following options is/are correct? $(a \in R)$.

A) Values of ' a ' for which $n(S_1 \cap S_2)$ is 1 is $\left\{\frac{-1}{\sqrt{2}}, 1+\sqrt{\frac{3}{2}}\right\}$

B) Values of ' a ' for which $n(S_1 \cap S_2)$ is 4 is $\left(1-\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)-\{0\}$

C) Number of values of ' a ' for which $n(S_1 \cap S_2)$ is 3 is 4 .

D) Number of values of ' a ' for which $n(S_1 \cap S_2)$ is 3 is 3 .

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; ZeroMarks : 0 In all other cases..

8. For any real number x , let [x] denotes the largest integer less than or equal to x and let r be a real-valued function defined on the interval [-20,20] by

$$f(x)=\begin{cases}\left[x+\frac{1}{2}\right]-x, & \text{if } \left[x+\frac{1}{2}\right] \text{ is odd} \\ x-\left[x+\frac{1}{2}\right], & \text{if } \left[x+\frac{1}{2}\right] \text{ is even}\end{cases}$$

then the value of $\frac{\pi^2}{5}\int_{-20}^{20} f(x)\cos(\pi x)dx$ .....

9. Number of complex numbers " z " satisfying $|z(3-4i)+\bar{z}(3+4i)+4|+|z(4-3i)+\bar{z}(4+3i)+6|=10$ and $|z+1|+|z-i|=\sqrt{2}$ is .....

10. $P(x)=0$ is a cubic polynomial equation with integral coefficients that has atleast one integral root. If $P(2)=5$ and $P(4)=3$ then sum of all possible integral roots is .....

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11. The sequence $a_1, a_2, a_3 \ldots\ldots\ldots$ satisfies $a_1 = 19, a_9 = 99$ and $\forall n \geq 3, a_n$ is the arithmetic mean of first ($n-1$) terms then a_2 is equal to……

12. If a, b are two positive integers such that $\dfrac{(a+b)^7 - a^7 - b^7}{ab(a+b)\left(a^2+ab+b^2\right)^2} = 7$ and $\sum_{r=1}^{6} {}^7c_r a^{7-r} b^r$ is divisible by 7^4 where none of $a, b, a+b$ is divisible by 7, then least possible value of $a+b$ is ___

13. Value of $\left(\tan\dfrac{\pi}{25}\right)\left(\tan\dfrac{2\pi}{25}\right)\left(\tan\dfrac{3\pi}{25}\right).....\left(\tan\dfrac{12\pi}{25}\right) =$ ____

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

$y = f(x)$ differentiable function $\forall\, x \in R$ and satisfies $\dfrac{dy}{dx} = |y| + 1$ and $f(0) = 0$

14. Number of integers in the range of f(x) in interval $x \in [-1, 1]$ is ……

15. If $a_1, a_2, a_3, a_4, ..., a_n$ are the solutions of equation $f(x) = \sin 2x$, then the value of $\left|[a_1] + [a_2] + [a_3] + ... + [a_n]\right|$ is (where [.] denotes greatest integer function)……

PARAGRAPH-II

Let n be the number of ordered 4-tuples (a, b, c, d) in which $(2026)^2$ is LCM of positive integers a, b, c, d

16. The digit in the units places of n is……

17. Let n_1 be the number of ordered 4-tuples (a_1, b_1, c_1, d_1) in which n is LCM of positive integers a_1, b_1, c_1, d_1 and n_r be the number of ordered 4-tuples (a_r, b_r, c_r, d_r) in which n_{r-1} is LCM of positive integers a_r, b_r, c_r, d_r for $r = 2, 3.....$ The digit in the 1000's place of n_{10} is…..

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PHYSICS **Max Marks: 60**

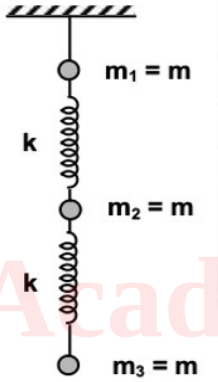
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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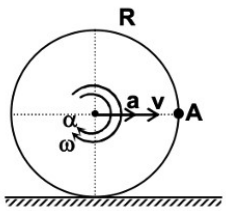
Full Marks : +3 If **ONLY** the correct option is chosen;
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18. Three identical balls of equal mass (m) are suspended vertically from a thread and two springs of same spring constant k as shown in the figure. The distance between the first and second ball and the second and third ball are same at equilibrium. The centre of mass of the whole system coincides with the second ball. The thread supporting the upper ball is cut and the system starts to fall freely. At some later time, it is observed that both the springs acquire their natural lengths. (Assume this happens) Find the distance of the centre of mass of the system from the second ball at that instant.



A) $\left(\sqrt{3}-1\right)\frac{mg}{2k}$ **B)** $\frac{mg}{3k}$ **C)** $\frac{mg}{2k}$ **D)** zero

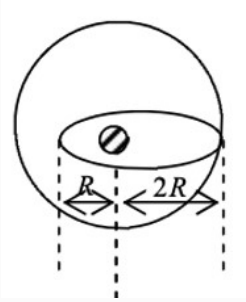
19. A uniform circular disc of radius $'R'$ is rolling without slipping on a rough horizontal surface with a constant acceleration $'a'$. Then the radius of curvature of trajectory of point $'A'$ of the disc relative to the ground at the given instant as shown in the figure is



A) 4R **B)** $2\sqrt{2}R$ **C)** 2R **D)** $\sqrt{2R}$

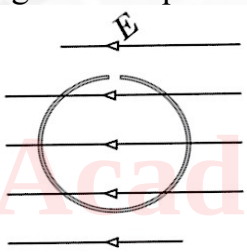
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20. A spaceship (which has negligible mass) is in an elliptical orbit about a planet. The minimum distance between the spaceship and the planet is R. At the point of maximum distance, spaceship is travelling at a speed v_0; It then fires his thrusters so that it enters a circular orbit of radius $2R$. What is its new speed:



A) $\frac{\sqrt{3}}{\sqrt{2}}v_0$ **B)** $\sqrt{5}v_0$ **C)** $\frac{\sqrt{3}v_0}{5}$ **D)** $\sqrt{2}v_0$

21. A thin rigid insulating ring of radius r and mass m has a very small gap of length $l(l << r)$ and carries a uniformly distributed charge q. Consider the mass density of the ring to be uniform. The ring is at rest in free space and a uniform electric field E in the plane of the ring and parallel to the gap is switched on as shown in the figure. The maximum angular speed of the ring in subsequent motion is____



A) $\sqrt{\frac{qlE}{\pi mr^2}}$ **B)** $\sqrt{\frac{qlE}{2\pi mr^2}}$ **C)** Zero **D)** $\sqrt{\frac{2qlE}{\pi mr^2}}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

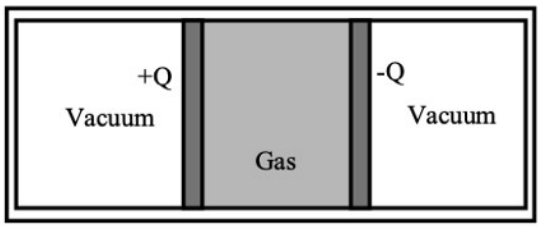
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22. A ideal gas having f degrees of freedom is kept inside a thermally insulated vessel, the vessel has two large pistons each with charge of $+Q$ and $-Q$. Assume charges are uniformly distributed. Pistons can move without friction inside the vessel. At an instant the charge of pistons is increased K times with the help of an external agent. Then which of the following statements are correct



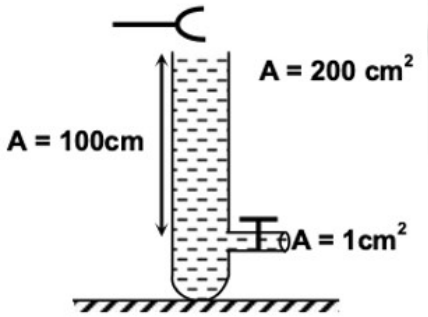
A) The Ratio of final to initial temperature of gas in equilibrium is $\frac{K^2+f}{2+f}$.

B) The Ratio of final to initial temperature of gas in equilibrium is $\frac{2K^2+f}{2+f}$.

C) Work done by the agent in the process is $\left(K^2-1\right)nRT_1$ where T_1 is initial temperature.

D) Work done by the agent in the process is $\left(K^2-1\right)\frac{Q^2L_1}{2A\varepsilon_0}$ where L_1 is initial separation between pistons.

23. A tuning fork of frequency 1000 Hz is sounded over Resonance tube as shown in the figure. Speed of sound is $v=320\,m/s$ and area of cross-section of tube is $200\,cm^2$ at the level of 100 cm a water outlet is attached. The water outlet has area of crosssection $a=1cm^2$ water outlet is opened at time $t=0$. Neglect the end effect of resonance tube.



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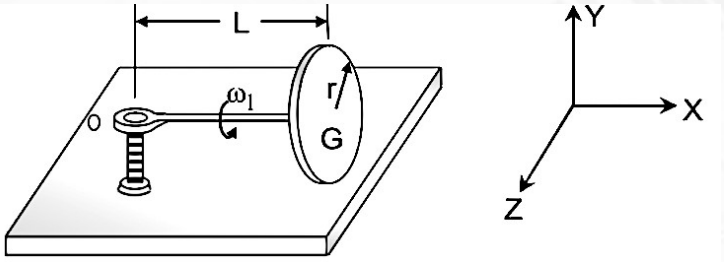
A) First resonance will occur when water level goes down by 8 cm from open end of tube.

B) First resonance will occur when water level is 16 cm below open end tube

C) The time after which resonance occurs first time is between 3 to 4 seconds.

D) The time between first two resonances is between 7 to 9 seconds.

24. A homogeneous disk of radius r and mass m is mounted on an axle OG of length L and negligible mass. The axle is pivoted at the fixed point O, and the disk is constrained to roll on a horizontal floor. Knowing that the 'disk' rotates counterclockwise at the rate ω_1 about the axle OG,



Choose the **Correct** option(s)

A) The angular velocity of the disk $\omega_1\hat{i} - \frac{\omega_1 r}{L}\hat{j}$

B) The angular velocity of the disc is $-\left(\frac{\omega_1 r}{L}\right)\hat{j}$

C) The X component of angular momentum of disc about 'O' is $\frac{1}{2}mr^2\omega_1$

D) The Z component of angular momentum of disc about 'O' is zero

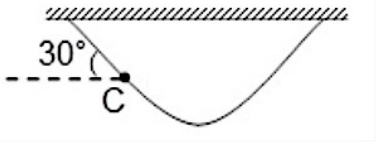
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

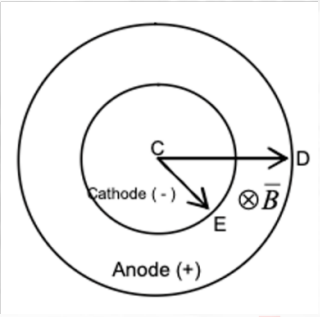
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25. A rope is hanging vertically, with its two ends fixed to walls at same horizontal level (as shown). Tension at the bottom most point is $90\,N$. If the mass/unit length of rope is $4\,kg/m$, find radius of curvature (in metres) at point 'C'. [Take $g = 10\,m/s^2$]

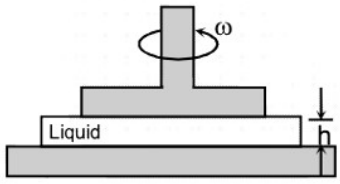


26. The electrodes of a diode are coaxial cylinders of radii inner $2m$ and outer $4m$. A potential difference V is maintained between them and their common axis is parallel to a uniform magnetic field $B = 1$ Tesla. The inner cylinder is the cathode, electrons leave it radially with negligible speed. If they will reach the anode at grazing incidence if $v = \frac{ne}{8m}$, then $n = ?$



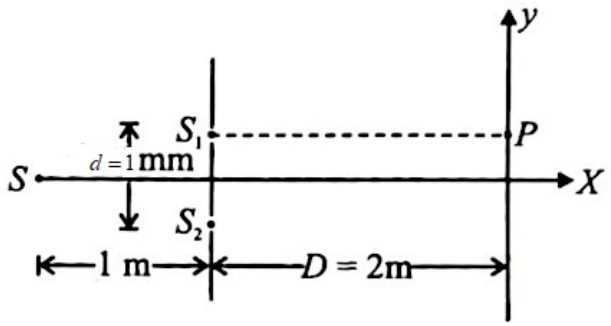
27. A free electron with kinetic energy $3.4\,eV$ happens to fall into an orbit around a fixed alpha particle forming a He^+ atom in the first excited state. A photon of wavelength λ_1 is emitted. Then the atom comes to the ground state emitting another photon of wavelength λ_2. If $\frac{\lambda_1}{\lambda_2} = \frac{12}{k}$, then find the value of k.

28. A circular disc of a diameter d is slowly rotated in a liquid of large viscosity μ at a small distance h from a fixed surface. An expression for torque τ necessary to maintain an angular velocity ω is $\frac{\mu\pi d^4\omega}{8Xh}$. Find the value of X.

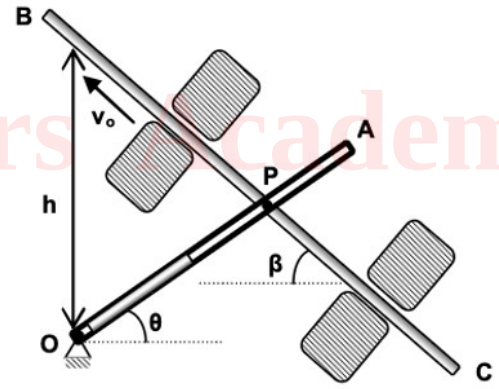


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29. In a modified YDSE the source S of wavelength $5000A^\circ$ oscillates about axis of setup according to the equation $y = 0.5\sin\left(\frac{\pi}{6}\right)t$, where y is in millimeter and t in second. At what minimum time t (in second) will the intensity at P, a point exactly in front of slit S_1, be maximum for the first time?



30. Pin P is attached to Rod BC and slides freely in the slot of OA, as shown in the figure. Determine the rate of change $d\theta/dt$ (in rad/s) of the angle θ at the instant when $h = 4m, \beta = 60^\circ,$ and $\theta = 30^\circ$, knowing that BC moves at a constant speed $v_0 = 10\,m/s$.



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SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
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PARAGRAPH-I

We aim to measure the resistivity of a homogeneous conducting material with the help of an ideal voltmeter and a current source in two different situations as shown in figure -1 and figure-2.

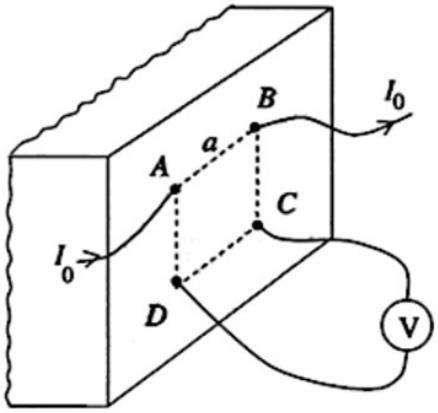


Figure-1

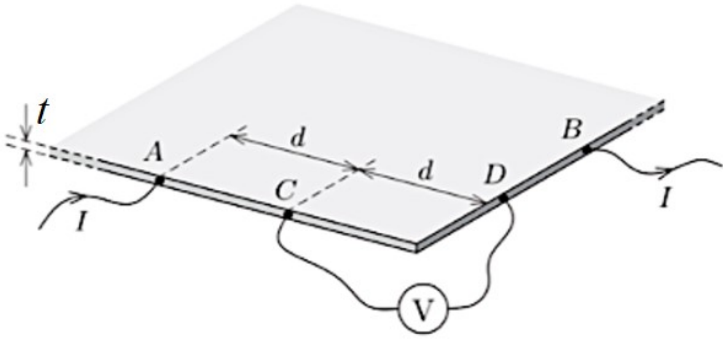


Figure-2

In Figure-1, a large plane divides space into two halves. One half is filled with a homogeneous conducting medium. We mark the outline of a square of side a on the plane and let a current I_0 in and out at two of its neighbouring corners A and B respectively. Meanwhile, We measure the potential difference V between the two other corners C and D.

In Figure-2, we aim to measure the resistivity of the material of a large, thin, homogeneous square metal plate, of which only one corner is accessible. To do this, we chose points A, B, C and D on the side edges of the plate that form the corner Points A and B are both $2d$ from the corner, whereas C and D are each a distance d from it. The length of the plate's sides is much greater than d, which, in turn, is much greater than the thickness t of the plate. If a current I enters the plate at point A, and leaves it at B, then the reading on a voltmeter connected between C and D is V. (Consider all connectors embedded into plates to be identical)

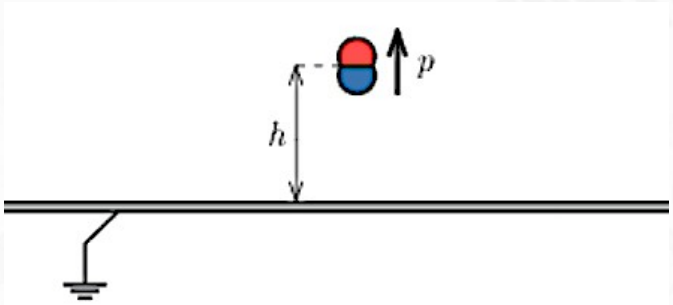
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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
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31. The resistivity of the material in case of Figure-1 is $\frac{n \times \pi a V}{I_0}$. Determine "$n$".

32. The resistivity of the material in case of Figure-2 is $\frac{\pi t}{2\ln(n/3)}\frac{V}{I}$. Determine"$n$".

PARAGRAPH-II

A small electric dipole is placed above a large, grounded, and horizontal metal plate at a height of h. This is done in such a way that its dipole moment p points upwards as shown in the figure.



33. Determine the distance (in metre)of the points on the metal sheet, from the dipole where the surface charge density is zero. $(h = 2m)$

34. The magnitude of the work done by the external agent in bringing the dipole from large distance to the given position slowly is $n \times \frac{p^2}{64\pi \epsilon_0 h^3}$. Determine "n".

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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. Choose incorrect statement about lanthanoids (Ln):

A) Lanthanoids form Ln_2C_3, LnC_2, Ln_3C types of carbides.

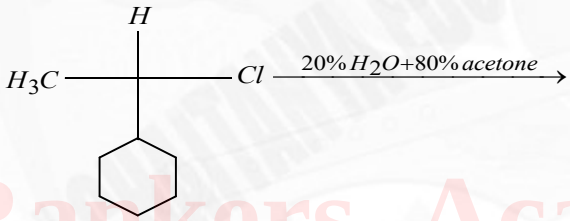
B) Absorption bands are narrow for tripositive hydrated lanthanoid cations probably because of excitation within f-levels.

C) Basic character of hydroxides of tripositive lanthanoid cation increases from Ce to Lu .

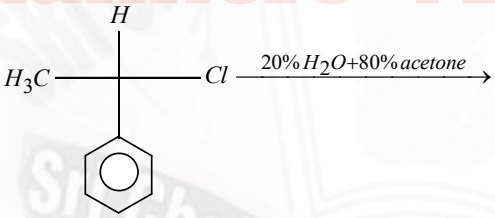
D) Lanthanoid ions are separated and purified by ion exchange method.

36. Among the following hydrolysis reactions, in which reaction percentage of racemisation is maximum?

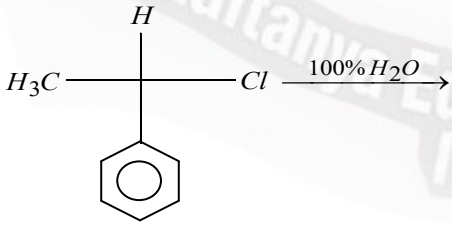
A)



B)



C)

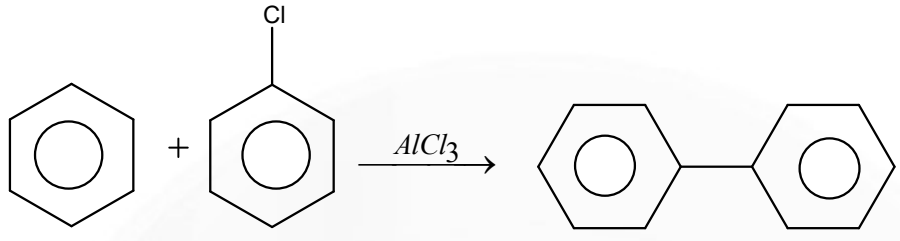


D) In the reactions b and c, percentage of racemisation is equal and greater than that of reaction a

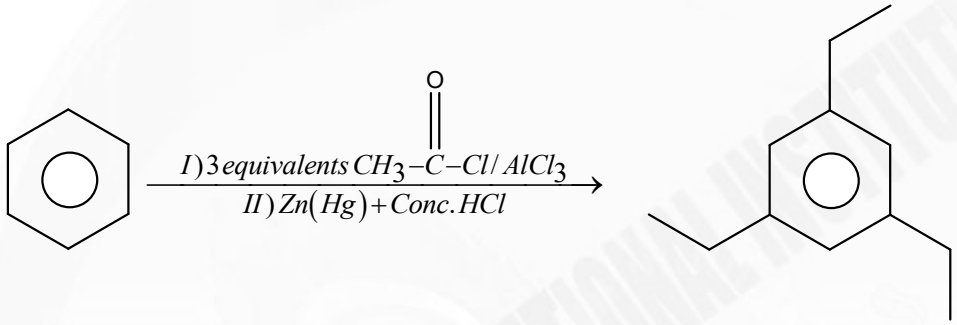
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37. In which one of the following reactions correct major product is given ?

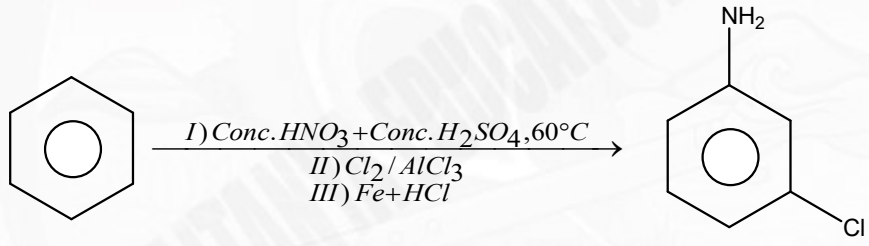
A)



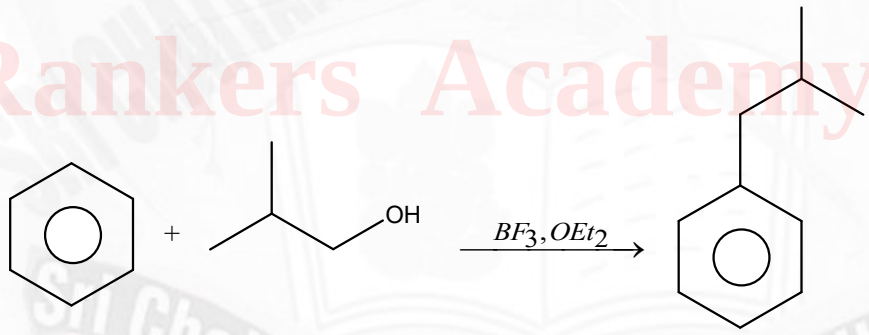
B)



C)



D)



38. A drug is known to be ineffective after it has decomposed to the extent of 48.8% (d.o.d = 0.488). The original concentration of a sample was 500 units/mL. When analysed 20 months later, the concentration was found to be 400 units/mL. Assuming that the decomposition is of first order, what will be the expiration time of the drug sample in months?

A) 58 **B)** 60 **C)** 62 **D)** 64

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Consider given reaction:

$$K_4\left[Fe(CN)_6\right] + KNO_2 + CH_3COOH \rightarrow X + CH_3COOK + Y(g)$$

If X is complex species and Y is gaseous substance then choose correct statement(s):

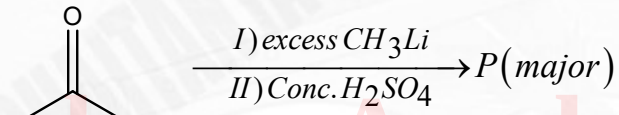
A) Both X and Y are coloured paramagnetic species.

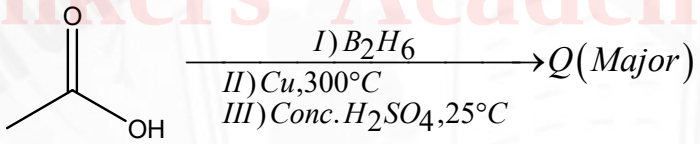
B) Both X and Y form coloured compound with $FeSO_4$ solution.

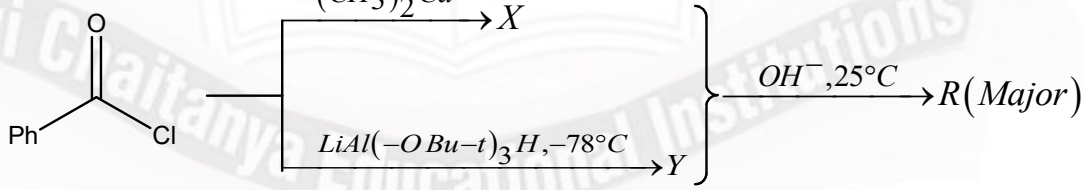
C) Y can be fluorinated by XeF_2.

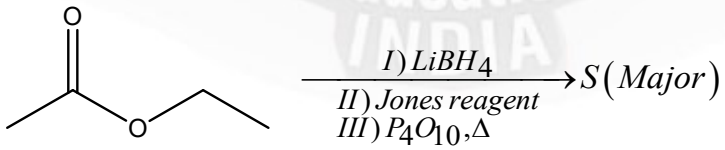
D) X produces blue colouration with KI solution in presence of dilute HCl and starch.

40. Four reactions are given below

R - I : 

R - II : 

R - III : 

R - IV : 

Among the following correct statement is/are

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A) 'P' gives coloured crystalline solid with Brady's reagent

B) 'Q' is heterocyclic aliphatic compound

C) Degree of unsaturation of 'R' is 9

D) 'S' can be used while preparing aspirin from salicylic acid

41. CO_2 gas is evolved, when solid Na_2CO_3 is

A) Fused with silica **B)** Boiled with $AgNO_3$ Solution

C) Treated with CH_3COOH **D)** Heating in Air

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; ZeroMarks : 0 In all other cases.

42. Consider oxidation of sodium borohydride by iodine in presence of diglyme:

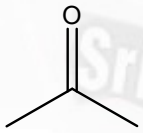
$$Na[BH_4] + I_2 \xrightarrow{\text{Diglyme}} NaI + P + Q$$

If Q is homonucleur diatomic molecule then how many moles of methanol is/are required for complete reaction with x moles of P obtained from two moles $Na[BH_4]$

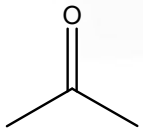
43. $$CH_3CHO \xrightarrow[II)\,H_3O^+ \\ III)\,\Delta]{I)\,HCN} X$$

'X' is a biodegradable polymer used for post operative stitches (Sutures). Gram formula weight (in g) of repeating unit of 'X' is

44.



1 mol + I_2 (1mol) $\xrightarrow{NaOH,\Delta}$ X (Organic compound with iodine)



1 mol + Br_2 (1mol) $\xrightarrow{CH_3COOH}$ Y (Organic compound with bromine)

Sum of maximum weights (in g) of X and Y formed is (to the nearest whole number)

(Gram atomic weight of H is 1, C is 12, Na is 23, Br is 80, I is 127 g)

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45. To the isobutane in $Conc.H_2SO_4$, isobutene is added slowly then a hydrocarbon 'X' is formed. X is a good fuel with high octane number

$$X \xrightarrow[monochlorination]{+Cl_2/hv} Y(isomers) \xrightarrow{Fractional\ distillation} Z(organic\ fraction)$$

Product Y and Z is

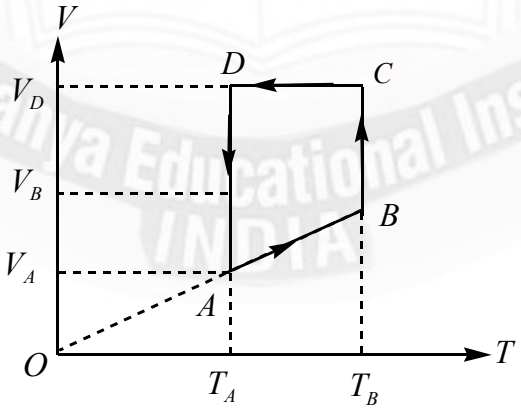
46. Calculate the amount (in milligrams) of SeO_3^{2-} in solution on the basis of following data. 20 mL of M/60 solution of $KBrO_3$ was added to a definite volume of SeO_3^{2-} solution. The bromine evolved was removed by boiling and excess of $KBrO_3$ was back titrated with 5 mL of M/25 solution of $NaAsO_2$. The reactions are given below.

[K - 39, Br - 80, As - 75, Na - 23, O - 16, Se - 78]

a) $SeO_3^{2-} + BrO_3^- + H^+ \rightarrow SeO_4^{2-} + Br_2 + H_2O$

b) $BrO_3^- + AsO_2^- + H_2O \rightarrow Br^- + AsO_4^{3-} + H^+$

47. A monoatomic ideal gas of two moles is taken through a cyclic process starting from A as shown in figure. The volume ratios are $\frac{V_B}{V_A} = 2$ and $\frac{V_D}{V_A} = 4$. If the temperature T_A at A is 27°C, calculate total heat absorbed by the gas in one cycle of the cyclic process (in Cal). $\left[C_{V_m} = \frac{3R}{2}\right]$



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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
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SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:*** **+3**If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Consider given chemical reaction: $H_3BO_3 + \underset{(\text{glucose})}{C_6H_{12}O_6} \rightarrow H_3O^+ + P$

and answer following questions.

48. How many moles of glucose molecules(s) is/are required for determining sharp end point for titration between three moles H_3BO_3 and NaOH solution in presence of phenolphthalein indicator.

49. Find out numerical value of expression $|x+y|$ for anionic species P

where, x = Number of five membered ring(s) present in P.

y = Number of oxygen atom(s) directly bonded to boron in P.

PARAGRAPH-II

In a lead storage battery, Pb (anode) and PbO_2 (cathode) are used. Concentrated D_2SO_4 (deuterated sulphuric acid) is used as electrolyte. The battery **initially** holds 2.5 L acid solution in it. In the discharge process, the density of solution fell from $1.40\,g/mL$ to $1.25\,g/mL$. The sulphuric acid of density $1.40\,g/mL$ is 50% by mass and that of density $1.25\,g/mL$ is 20% mass of solution.
[At wts : S-32, O-16, D-2]

50. The concentration of D_2SO_4 after this discharge process in molarity is

51. The amount of charge in Faradays which the battery must have delivered under ideal conditions is

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30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

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